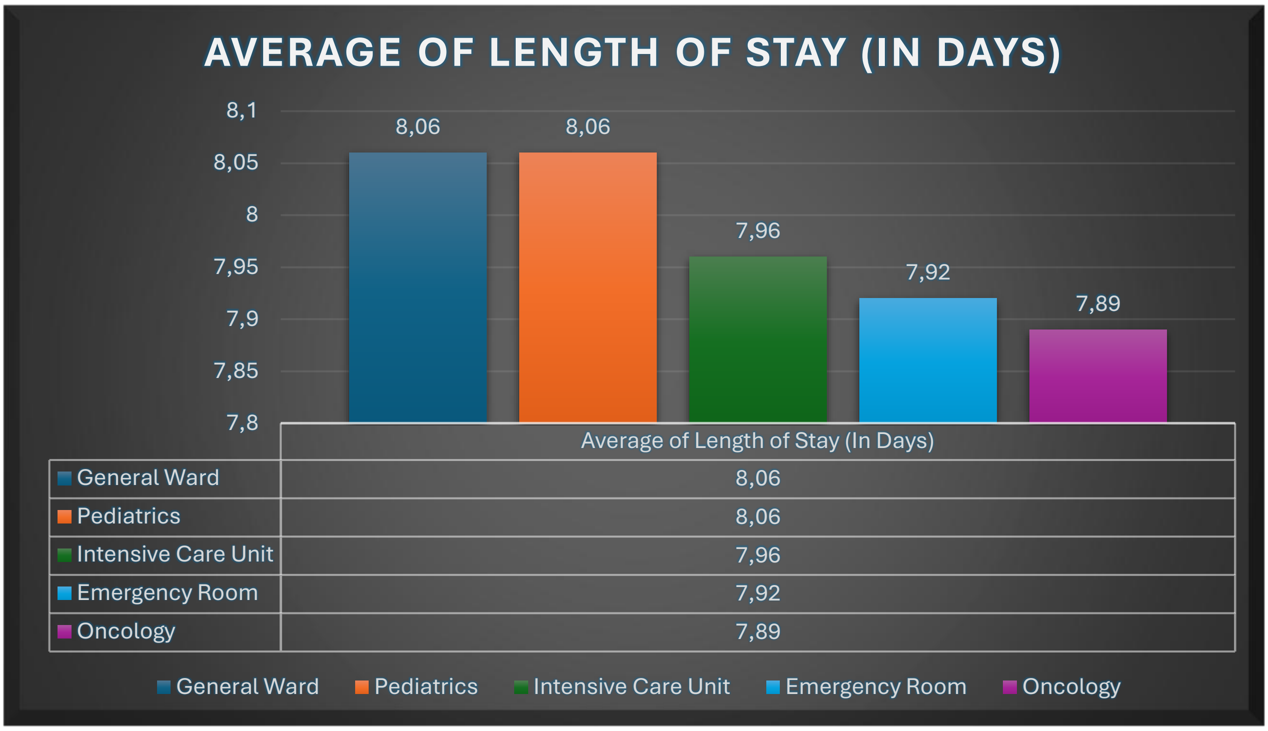
Task: Analyze hospital bed occupancy trends and patient flow efficiency using large-scale hospital data to identify bottlenecks and optimize resource allocation.

Okay, so using Microsoft Excel, I checked the data and I made a bit of a change. I set the titles for each of the columns to start with little letter – for example diagnosis_code, instead of Diagnosis_Code, better for machine learning.

Also I added the diagnosis with words after the it's code and the abbreviations have been changed to the full name – for example ER is Emergency Room.

Descriptive Analysis

Calculating average length of stay per department



(I am using Microsoft Excel to create this beautiful chart as visualization.)

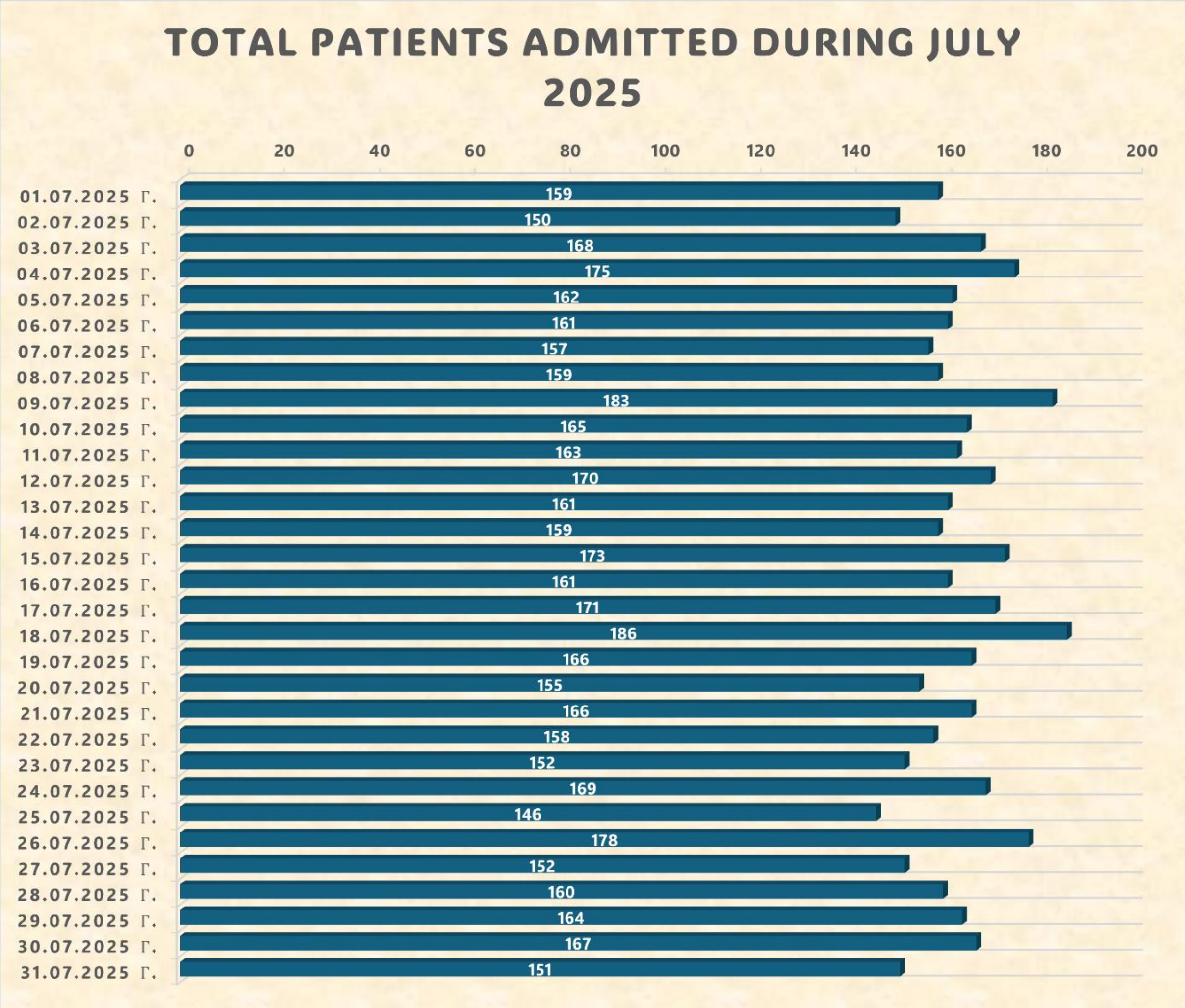
This is a visualization which shows the average length of stay for each of the departments.

- General Ward and Pediatrics are leading with the time – **8 days + 1 hour and 44 minutes for each;**
- Followed by Intensive Care Unit – **7 days + 23 hours, 2 minutes and 24 seconds;**
- Next is Emergency Room – **7 days + 22 hours, 4 minutes and 4 seconds;**
- And final – Oncology – **7 days + 21 hours and 6 minutes.**

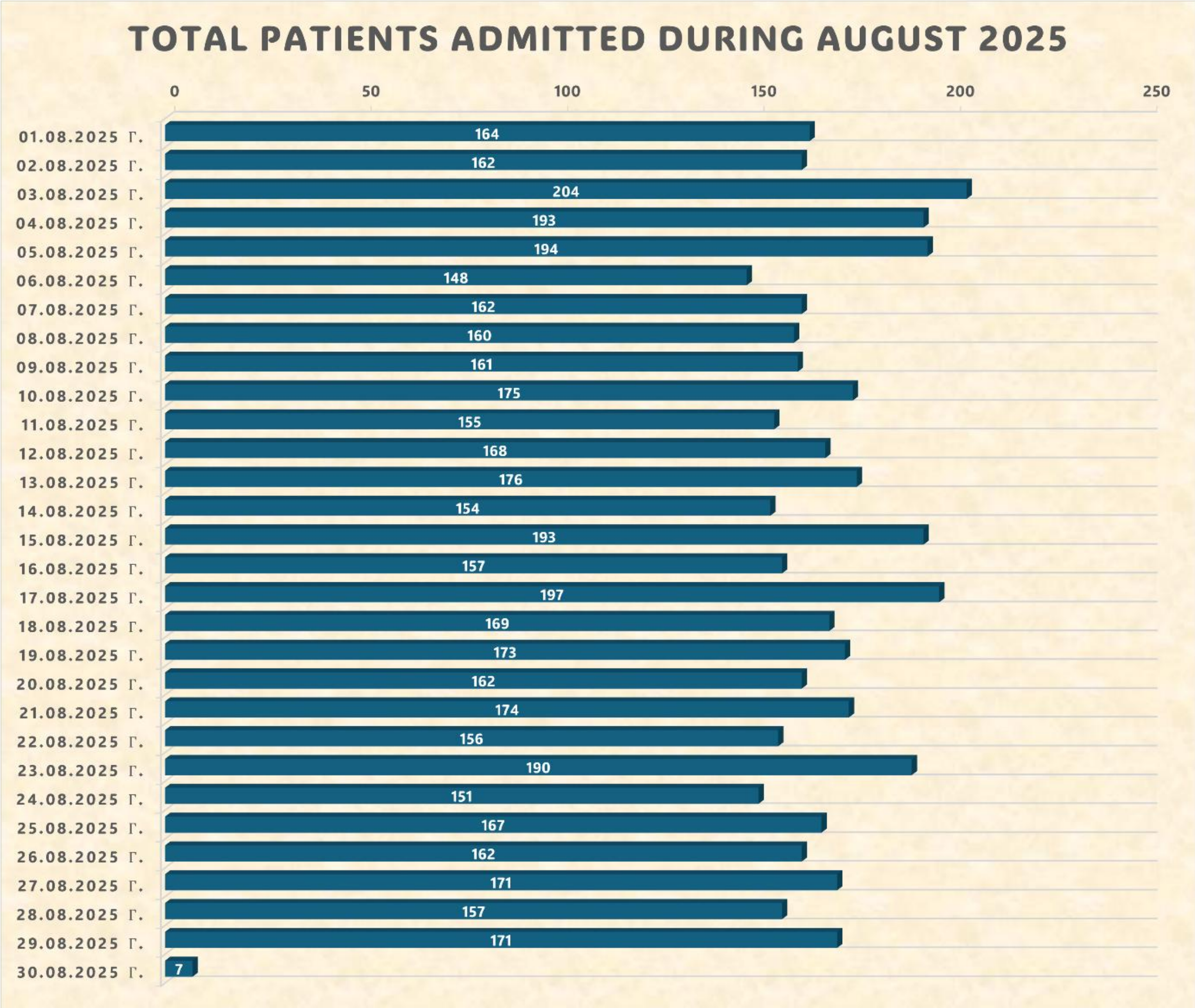
Recommendations:

- Better medical equipment for better quality work, if there isn't any;
- More medical workers, if there are not enough;
- More and/or with better quality training programs;
- More days off, if the medics are getting so much tired (keep in mind that they are people as well, not robots and they need to rest also).

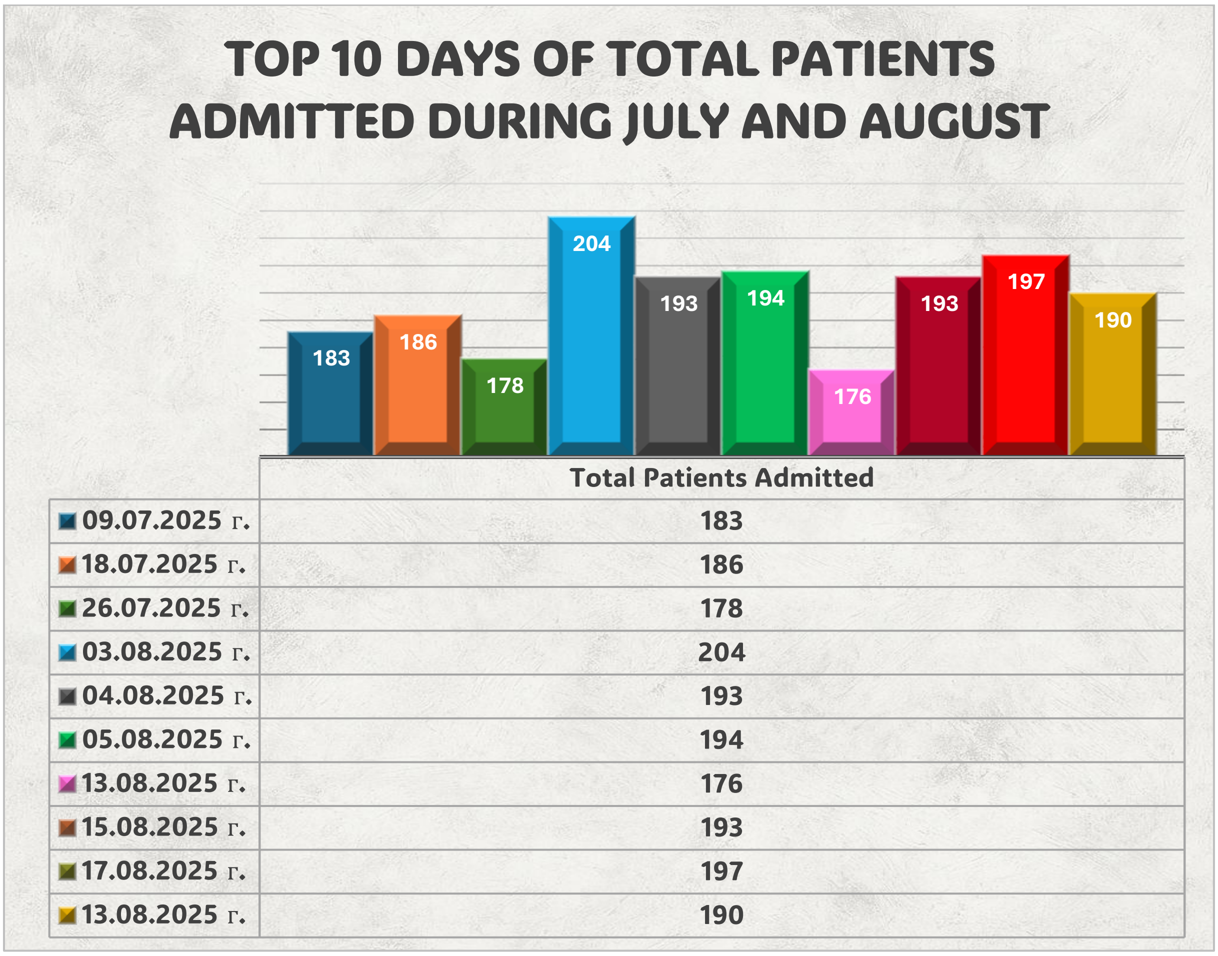
Identifying peak admission hours/days



This chart visualizes the daily number of patients admitted throughout July 2025. It highlights fluctuations in hospital activity, helping identify peak that may impact staffing, resource allocation, and operational planning. This data-driven insight helps for better decision-making and strategic improvements.



And here is for August as well. It looks like that perhaps September is starting with greatly less patients.



This is a chart, showing the dates with the most admitted patients during the summer season. Each of these days are so much people accepted in the hospitals.

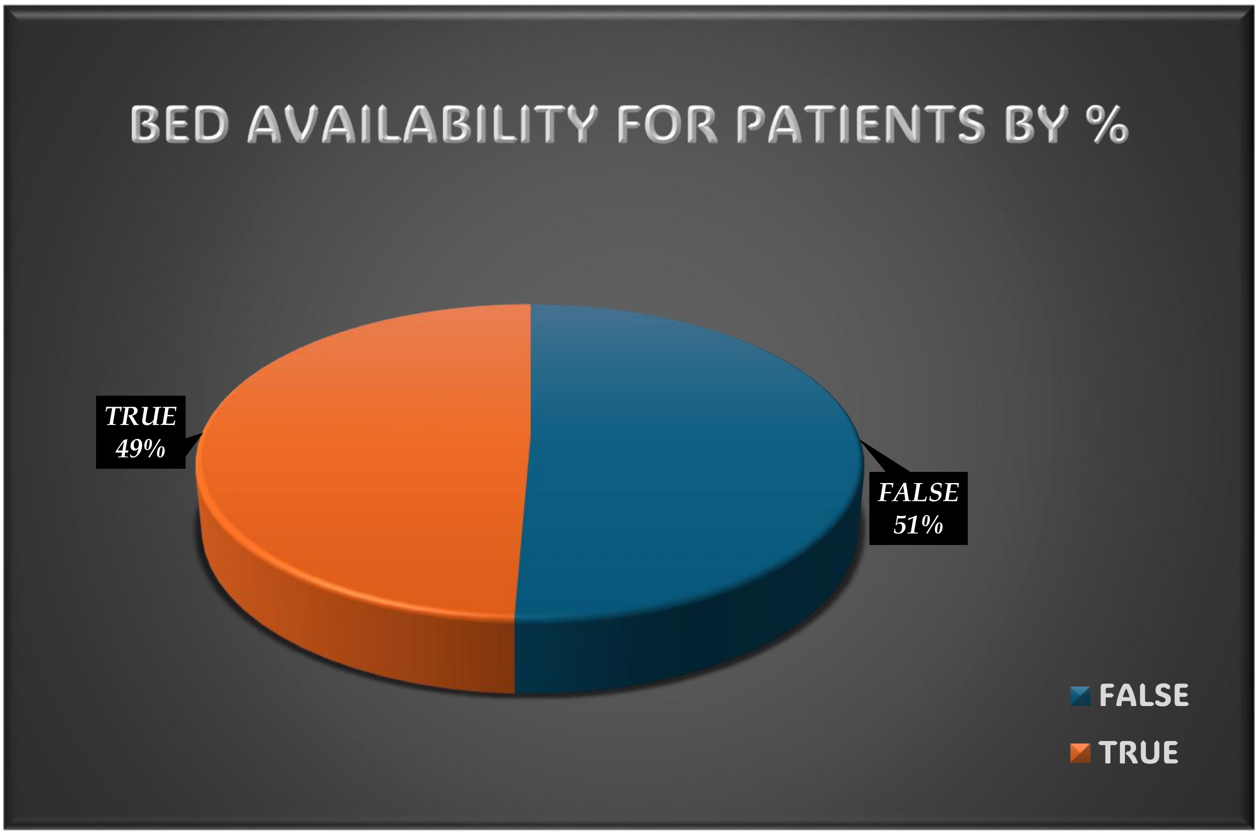
Recommendation: There are so many patients during the summer. There may be a virus or a flu at the hot time of the year. I recommend to remake the working schedule for more medics to be on shift or a bit more time with higher salary. Let us not forget, they are people as well and doing their best to keep up the sick people.

Determining bed occupancy rates over time

I am going to use a pivot table, to find out is there many available beds or perhaps not for most of the patients:

Using bed_availability in rows section and in values section patient_id and set to count – I am receiving the result:

Bed Availability	Percent of Patients
FALSE	50,66%
TRUE	49,34%
Grand Total	**100,00%**



This chart shows it well enough.

For all the patients in the dataset I am analyzing – looks like that there was no available bed for 51% of them, which is a high percent. I will directly recommend with no delays:

The hospital <u>must</u> get more beds. It is so important, because it is about healing people. The medical workers are focusing on the patients to keep them alive and healthy, no matter what the diagnosis is.

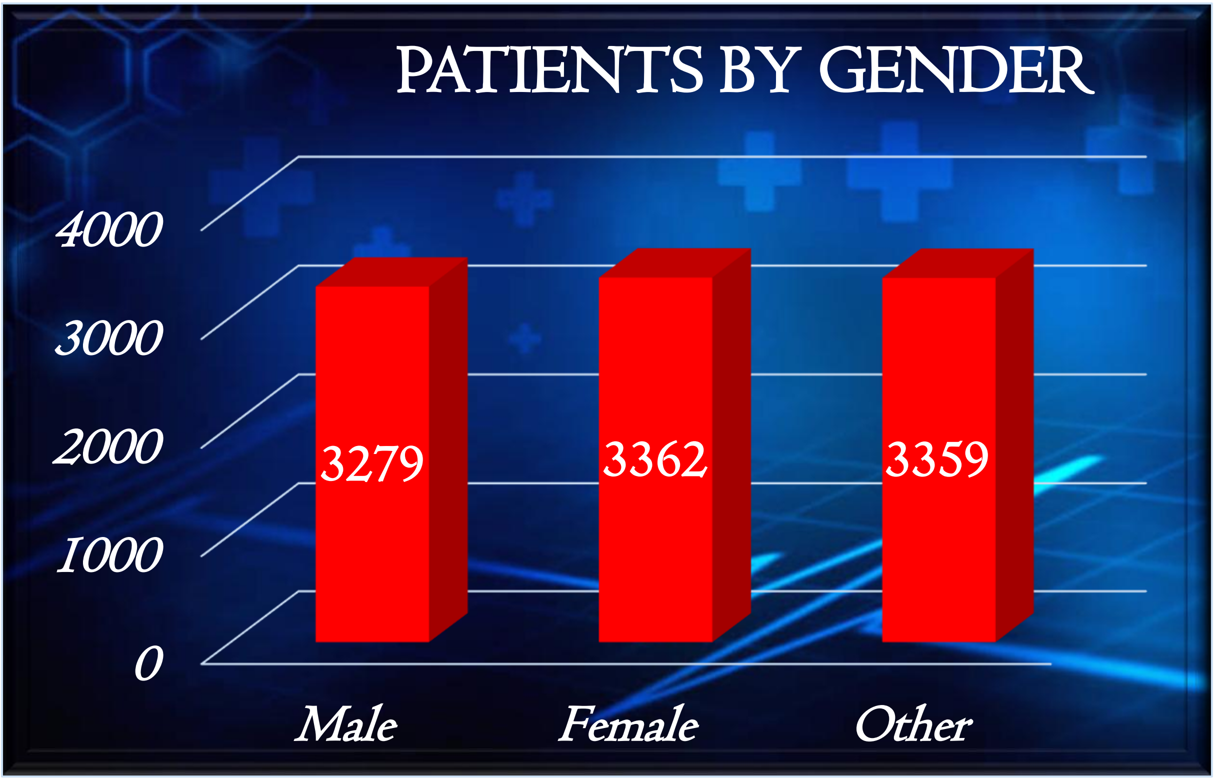
TREND DETECTIONS

I am starting to look for interesting situations that may greatly help some people:

First, let me check the genders – more men or more women as patients:

Here is my pivot table:

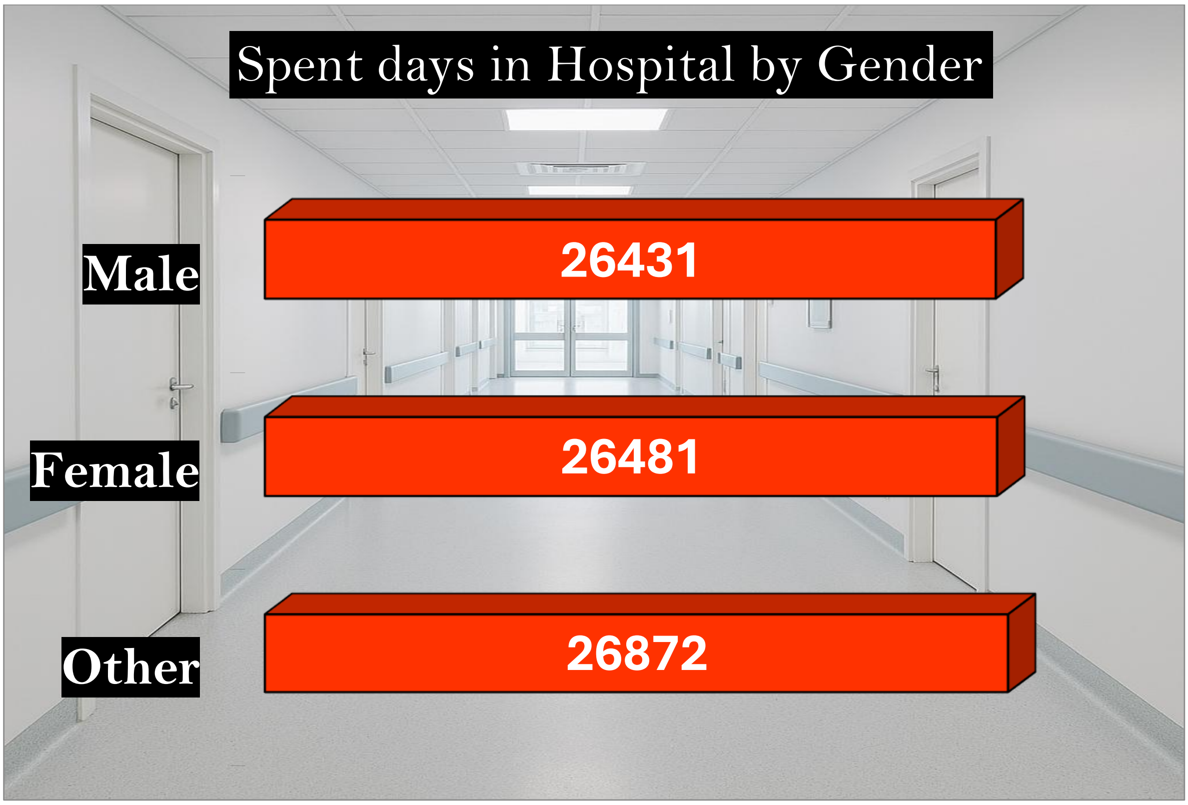
Row Labels	Count of patient_ID
Female	3362
Male	3279
Other	3359
Grand Total	**10000**



There isn't much of a difference, but there are people with unknown gender. Perhaps they do not want to share about that or perhaps it is possible to be a third gender. I do not know for sure, but this is what I have as data. Females are a bit more than the males and the unknowns. A possible reason to be that women are more than the men???

Next, I will check the by gender – men or women spend more time in hospital:

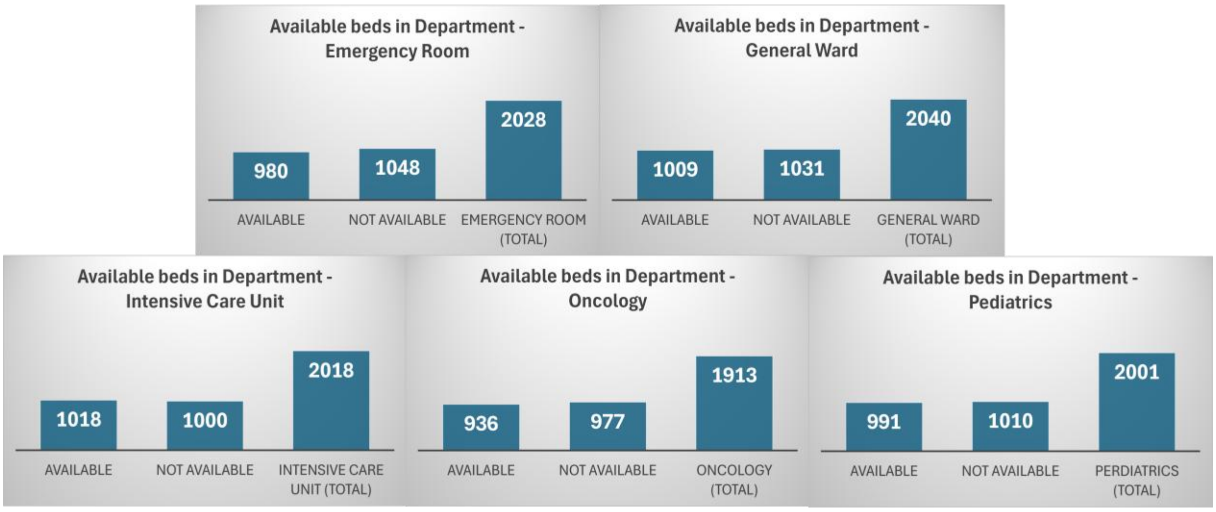
Row Labels	Sum of length_of_stay
Male	26431
Female	26481
Other	26872
Grand Total	**79784**



There isn't big difference. It is important all patients to receive care for their health white in the hospital, because when there are real professionals in that field, then people will remember them as good people and real health caretakers.

Departments with frequent overcapacity issues

I will check now how is the situation with the departments, are there enough beds for all the patients or not quite:



Emergency Room – most of the beds were not available;
General Ward – most of the beds were not available;
Intensive Care Unit – most of the beds were available;
Oncology – most of the beds were not available;
Pediatrics – most of the beds were not available

Recommendation: There are many beds unavailable in each of the departments. The healing of the people is so much important. The hospitals must invest in more beds!

Conclusion

The healthcare is the most important field, every person in this world receives problems with their health. The healers working in the hospitals are actually a second change to all people, so it is very valuable the presence of a strategy that will do everything that is possible for all of us to be alive and healthy.

I would say that we should position the healthcare field in first place, so no one must underestimate all medical workers that are educated and do everything they can to aid us, even if they need to work extraordinary.

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Used programs and platforms: Microsoft Word, Microsoft Excel